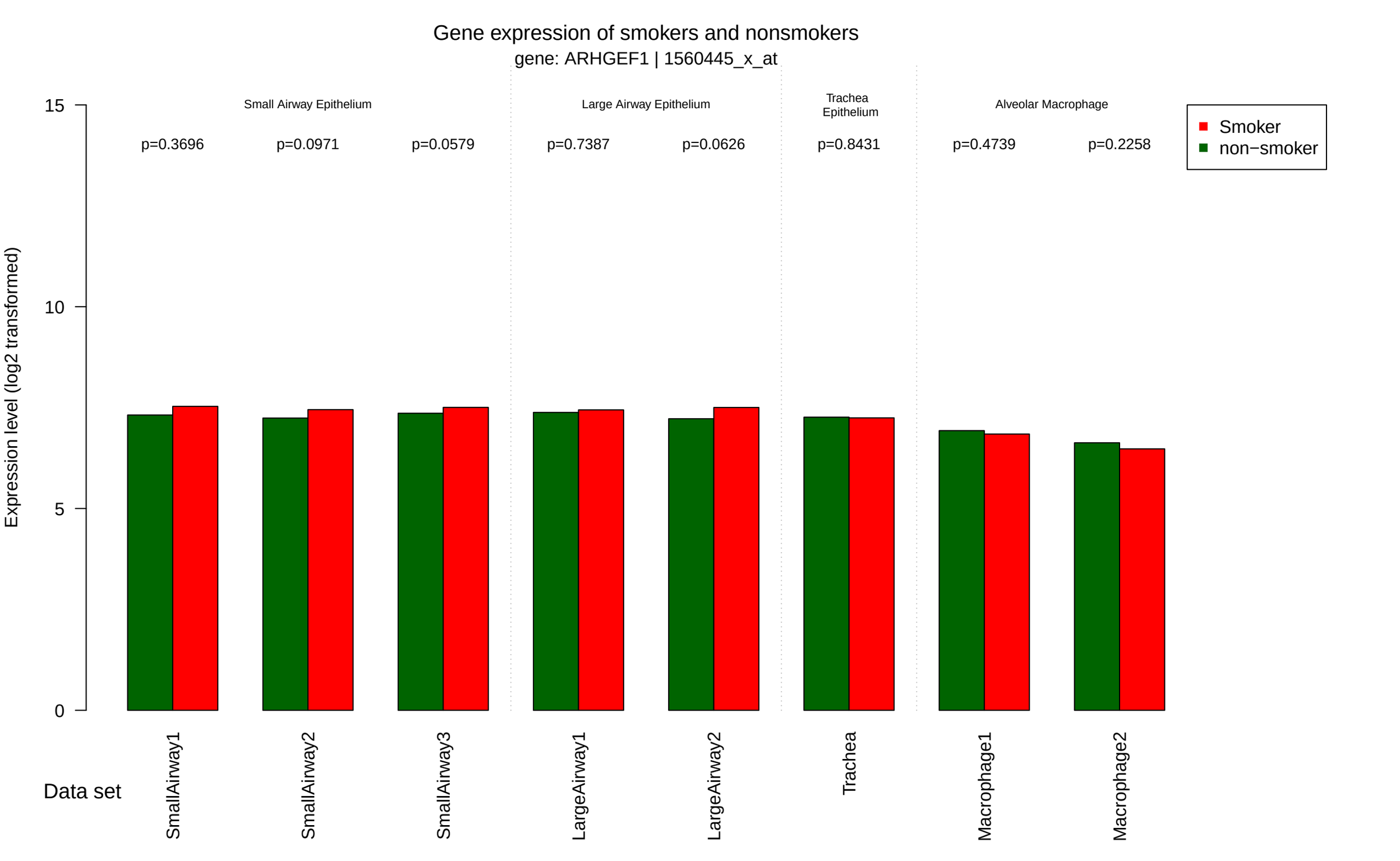
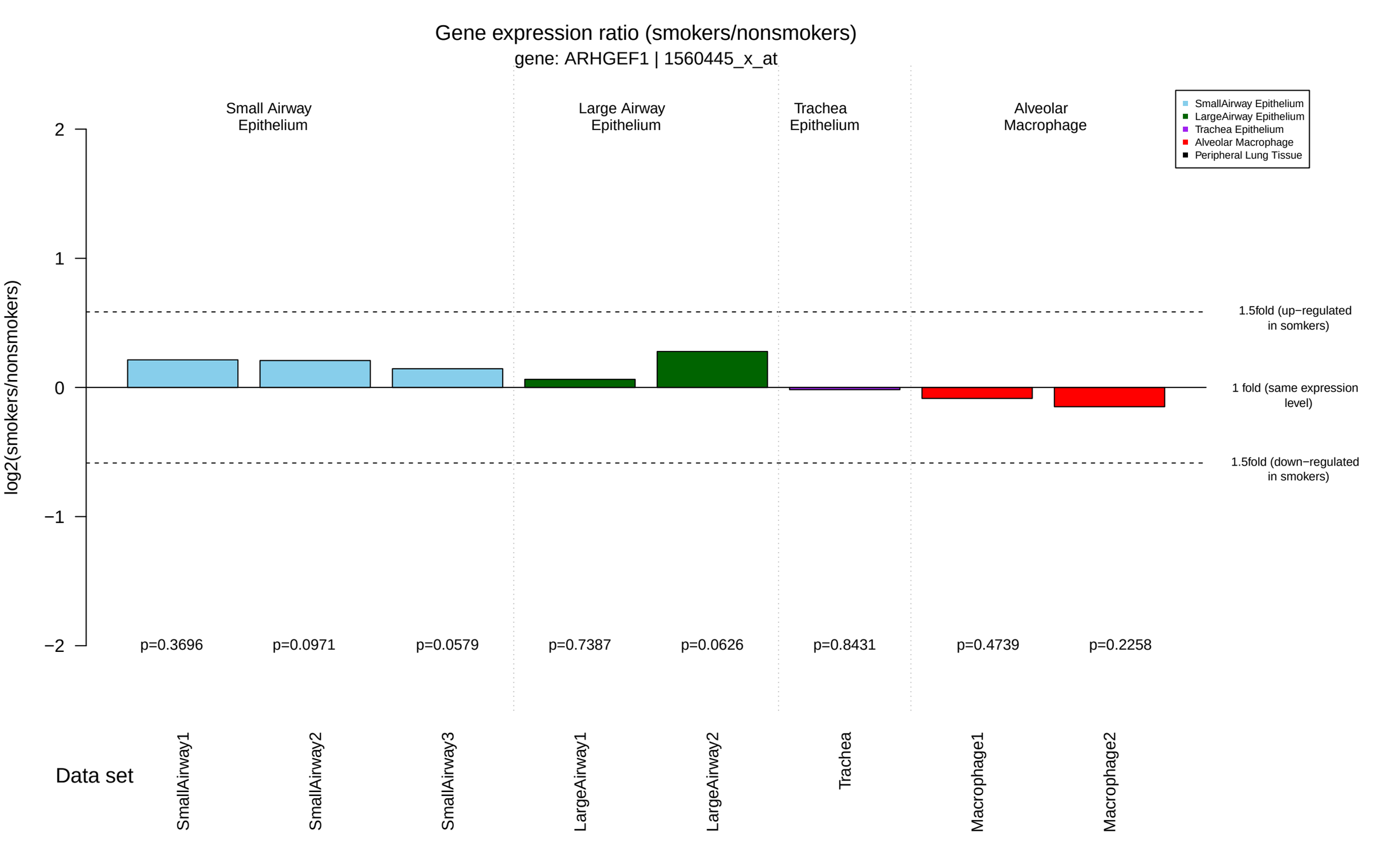
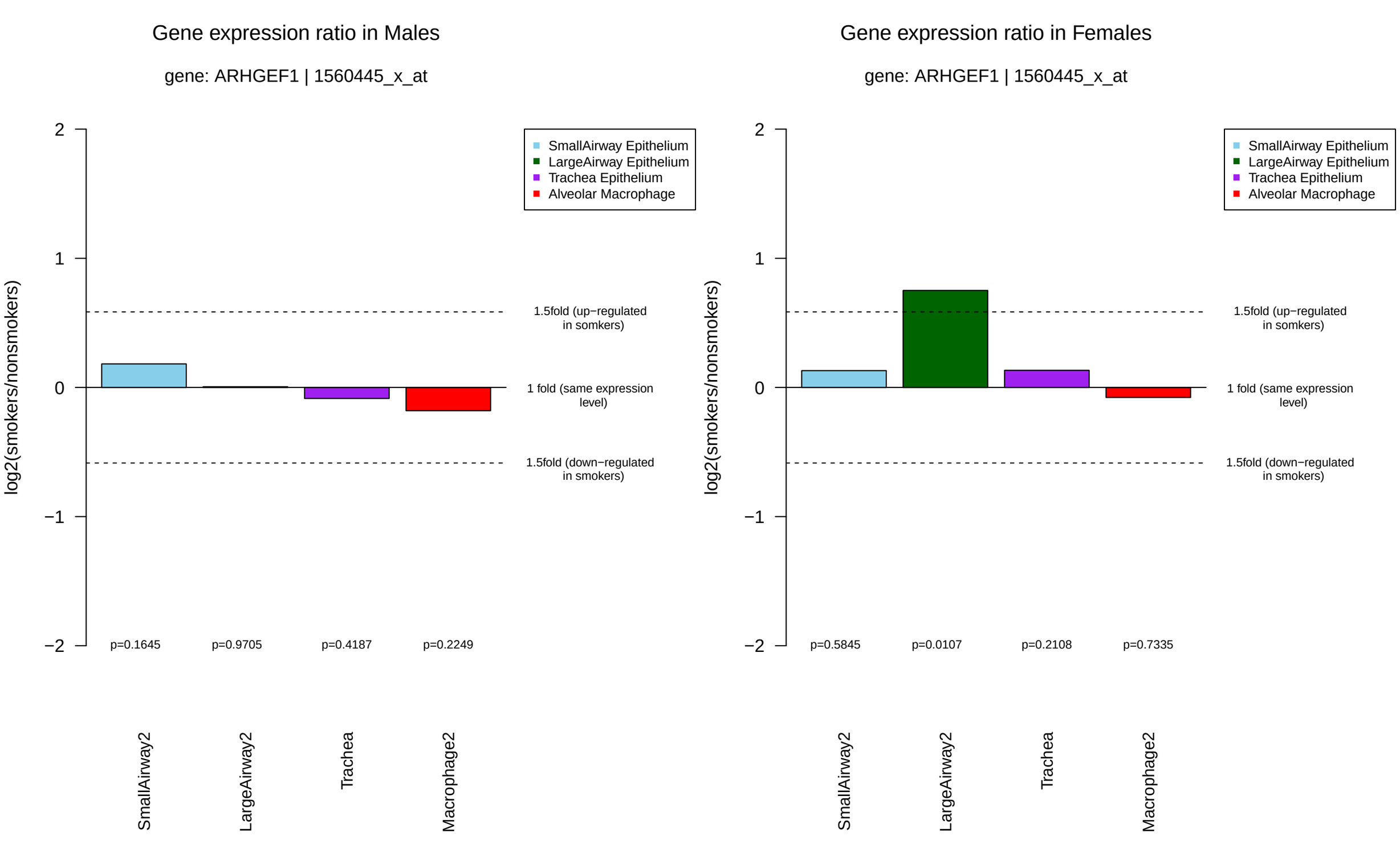
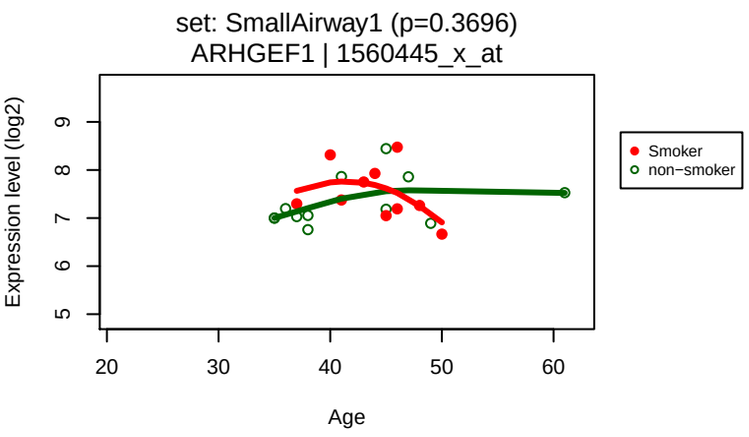
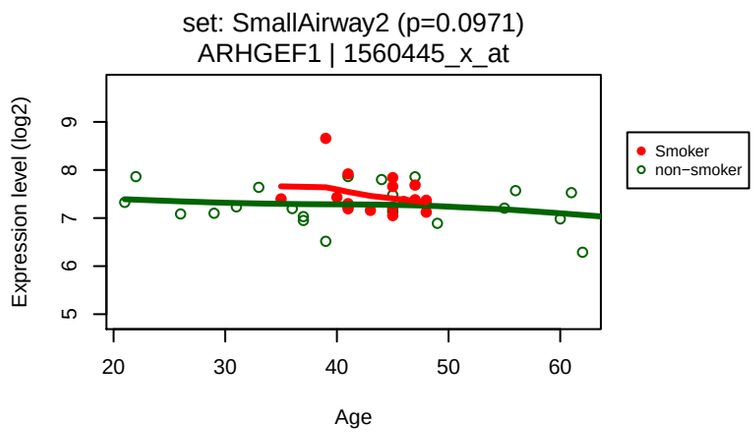
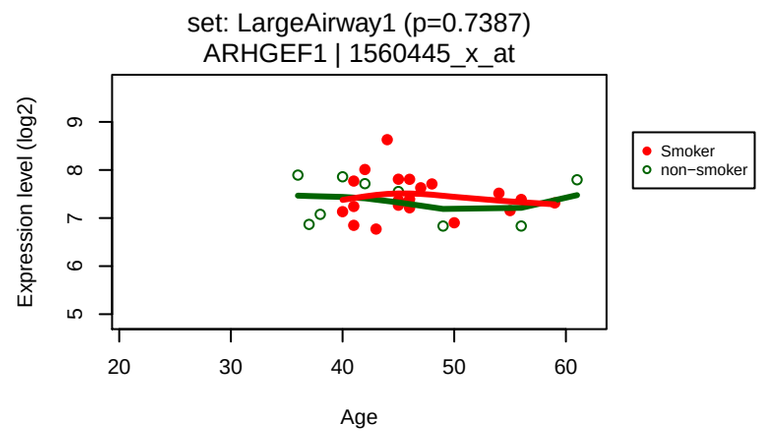

Gene expression of smokers and nonsmokers
gene: ARHGEF1 | 1560445_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3696
p=0.0971
p=0.0579
p=0.7387
p=0.0626
p=0.8431
p=0.4739
p=0.2258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

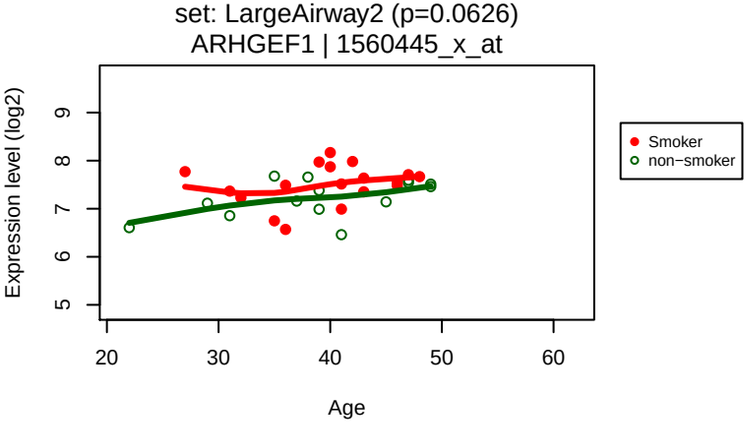

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF1 | 1560445_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3696
p=0.0971
p=0.0579
p=0.7387
p=0.0626
p=0.8431
p=0.4739
p=0.2258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

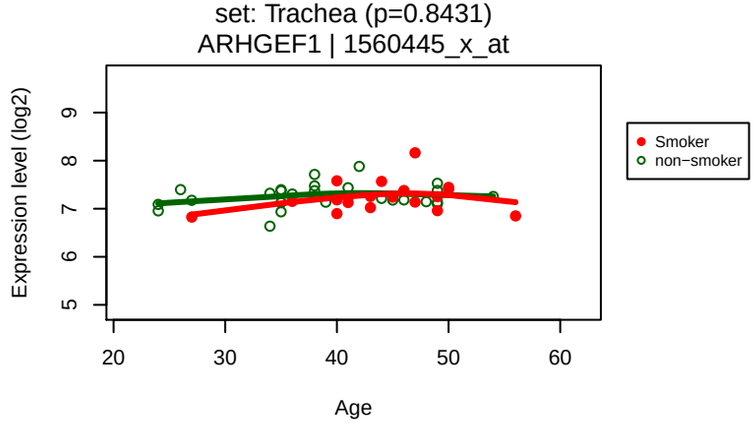

Gene expression ratio in Males
gene: ARHGEF1 | 1560445_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1645
p=0.9705
p=0.4187
p=0.2249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF1 | 1560445_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5845
p=0.0107
p=0.2108
p=0.7335
SmallAirway2
LargeAirway2
Trachea
Macrophage2

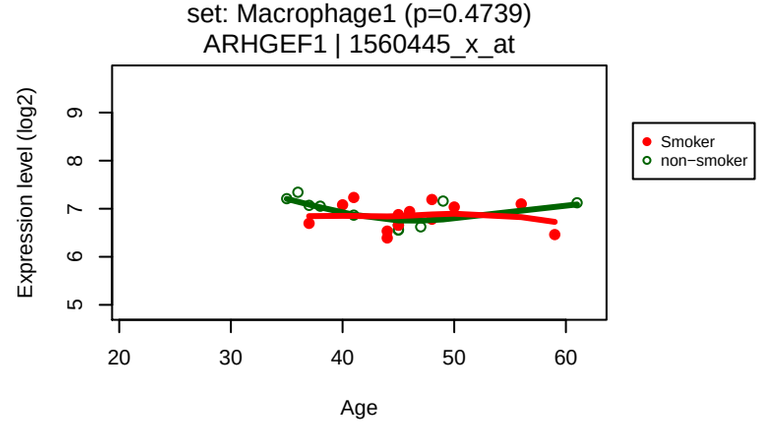

set: SmallAirway1 (p=0.3696)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker

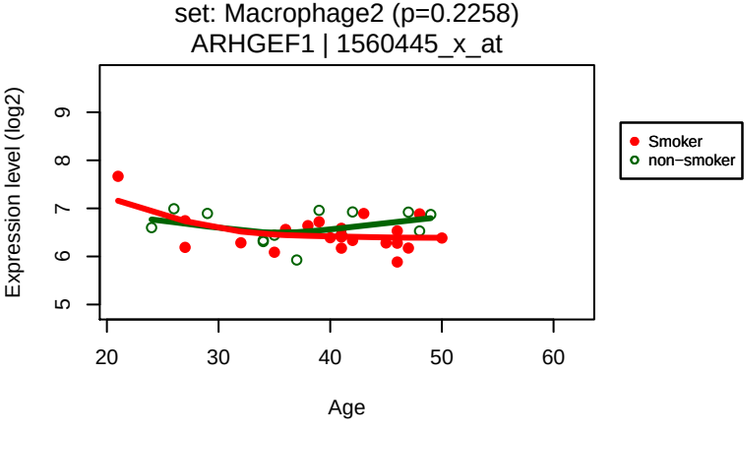

set: SmallAirway2 (p=0.0971)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker

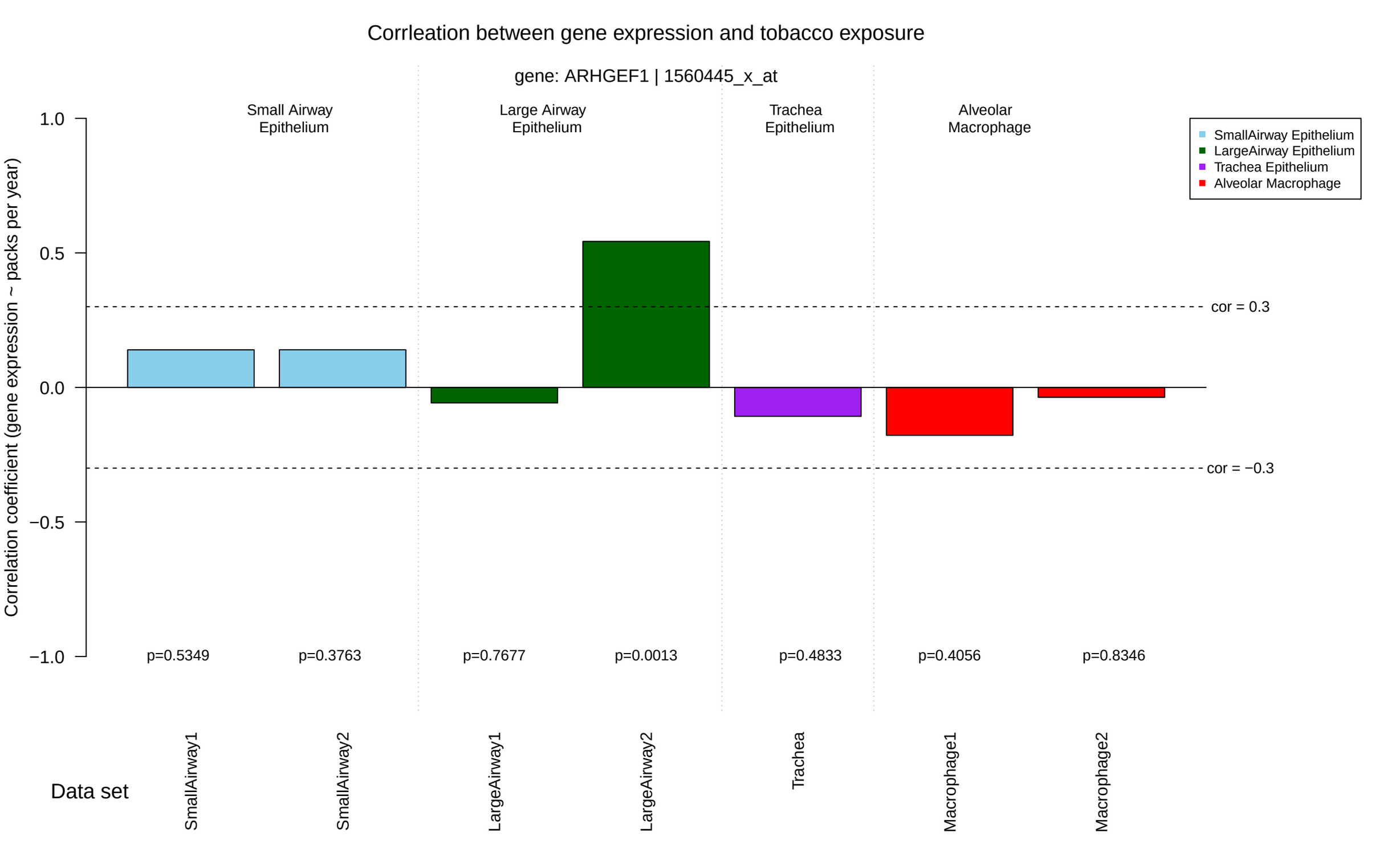

set: LargeAirway1 (p=0.7387)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0626)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8431)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4739)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2258)
ARHGEF1 | 1560445_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF1 | 1560445_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5349
p=0.3763
p=0.7677
p=0.0013
p=0.4833
p=0.4056
p=0.8346
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2